

北京邮电大学 2020—2021 学年第一学期

《数据库系统原理》期中考试试题

考试 注意 事项	一、学生参加考试须带学生证或学院证明，未带者不准进入考场。学生必须按照监考教师指定座位就坐。 二、书本、参考资料、书包等物品一律放到考场指定位置。 三、学生不得另行携带、使用稿纸，要遵守《北京邮电大学考场规则》，有考场违纪或作弊行为者，按相应规定严肃处理。 四、学生必须将答题内容做在试题答卷上，做在试题及草稿纸上一律无效。 五、填空题用英文答，中文答对得一半分。									
课程	数据库系统原理		考试时间		年 月 日					
题号	二	三	四	五	六	七	八	九	总分	
满分	6	5	24	25	40				100	
得分										
阅卷教师										

1. Fill in blanks. (1×6 points)

- (1) _____ data independence characteristic defines that when DBS's logical schemas at logical level (DBMS-oriented) change, application-oriented external schema or application programs do not change.
- (2) A _____ is a set of one or more attributes that taken collectively, allow us to identify a tuple in the relation, but there maybe the redundant attribute in it.
- (3) The _____ defines the specification of managing data items in database. It is a collection of conceptual tools for describing data structure, data relationships, data semantics, data operations and consistency constraints.
- (4) The _____ values in any tuple of relation r_i are either null or must appear as the primary key values of a tuple of the referenced relation r_j .
- (5) Database design involves the following phases: requirements analysis, conceptual schema design, _____ design and physical schema design.
- (6) Let the number of tuples for the relations R and S be 20 and 40, and the number of attributes be 3 and 4, respectively. The relationship T is a Cartesian product of R and S. Then, the number of tuples of T is _____ and the number of attributes is _____.

2. Choice (5 points)

- (1) The following SQL statement corresponds to the expression _____.

Select *

From r, s

- A. $r \cap s$ B. $r \cap s$ C. $r \times s$ D. $r - s$

- (2) In a Select statement, _____ can be used to take out repetition tuples.
 A. unique B. count C. distinct D. union

- (3) Aiming at security, the leader of each department can only access the files of the staff of his department, thus the DBA should create the corresponding _____.
 A. Index B. View
 C. Trigger D. Table

- (4) Given two relations $R(A, B, C, D)$ and $R2(B, D, E, F)$, in the following relational algebraic expressions, _____ is not right.
 A. $\Pi_{A,B}(R) \bowtie \Pi_{B,D}(R2)$ B. $R1 \cup R2$
 C. $R1 \bowtie R2$ D. $\pi_{A,B}(R1) \bowtie \Pi_{B,D}(R2)$

- (5) Given the following two relations:

R	A	B	C
1	b_1	c_1	
2	b_2	c_2	
3	b_1	c_1	

S	D	E	A
	d_1	e_1	1
	d_2	e_2	1
	d_3	e_1	2

Assume that R's primary key is A, S's primary key is D, and the foreign key clause is included in the definition of the relation S:

"FOREIGN KEY (A) REFERENCES R(A) ON DELETE RESTRICT".

- The following SQL statement which can not be successfully executed is _____.
 A. DELETE FROM R WHERE $A=2$
 B. DELETE FROM R WHERE $A=3$
 C. DELETE FROM S WHERE $A=1$
 D. DELETE FROM S WHERE $A=2$

3. (24 points) The schema diagram in the database University is given below (see Figure 1), and the primary keys in the relational tables are underlined. The table Department, Student, Teacher and Course describe the four data objects in database. A teacher in a department can teach a course in a class, and a student takes a class to learn a course, in the Spring or Fall Semester in a year.

The table Section maintains the information on all classes (sections) taught. Each section/class is identified by a course_id, sec_id, year, and semester, stating that a course is taught in a class/section in the Spring or Fall semester in a year. The location and time of the course offering in a section are given by building, room_no, and time_slot_id.

For the following queries, give SQL statements for (1)~(3), and relational algebra expressions for (4).

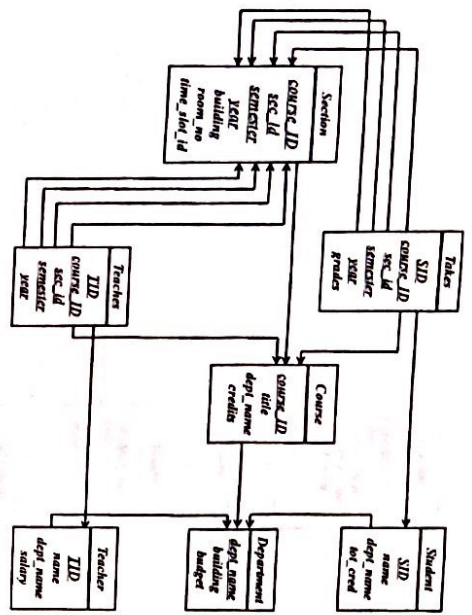


Figure 1. the schema diagram in the database University

- (1) Find the names and average salaries of all departments whose average salary is greater than 32000. (6 points)
- (2) Find the names of all students who are in IS department and have tot_cred values greater than that of all students in CE department. (6 points)

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- (3) For a student, if the total number of the courses he takes in the 2016 Spring semester is lower than 3, then decrease his tot_cred by 5 percent. (6 points)

- (4) Give a relational algebra expression, to find the TID and names of all teachers belonging to the departments located in Building4. (6 points)

4. (25 points) Convert the following E-R diagram in Figure 2 to the proper relation schemas and identify the primary key of each relation.

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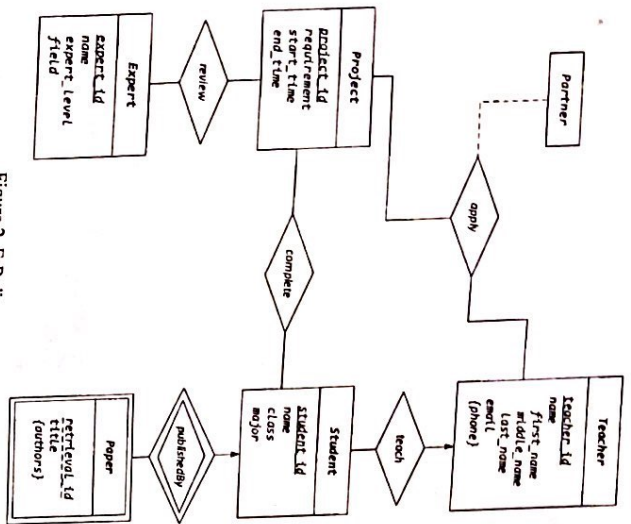


Figure 2. E-R diagram

5. (40 points) Consider the following information of the database of a school sport-meeting management system. The data requirements are summarized as follows:

- (1) Every **Competition category** (比赛类别) is identified by a **category_id** and has a name and a manager.
 - (2) **Competition events** (比赛项目) are identified by **event_id**. For each event, the name, competition time and level must be recorded.
 - (3) **Department teams** (系团队) are identified by **team_number**. Every team has a name, a leader.
 - (4) **Players** (运动员) are identified by **player_id**. For each player, the name, age, sex and phone number must be recorded.
 - (5) Every competition category has several competition events. Each event belongs to a unique category.
 - (6) Every team has several players. Each player belongs to a unique team.
 - (7) Each player could attend different competition events. And each event can be attended by more than one player. Players have their grades in different events.
1. (10 Pts) Draw an E-R diagram for the requirements above. In addition to the ER basics (entity sets, relationships and attributes), please identify the primary key of each entity set, and give the mapping cardinalities on the E-R diagram.

2. (10 Pts) Convert the E-R diagram to relational schemas, and also identify the primary key of each relation.

3. (4 Pts) Based on your relational schemas, write down the SQL statements to create the table Player. And also give the primary key and foreign key constraints.

4. (11 Pts) Write down SQL queries for the following:

(1) (5 Pts) Find name of player, name and time of Competition event for the team of "computer department team".

(2) (6 Pts) Find player_id and average competition grade of each player in the "computer department team", whose average grade of the competition is more than 85.